LESSON PLAN · SCRATCH JUNIOR (K–2) BEGINNER: LITTLE CODERS

Lesson 2: Numbers on Blocks

Module 1 | 30 minutes | Parameters (numbers on blocks) | KG–Grade 2 · ages 5–7

Objective: Change the number on a block to move more than one square, and choose the fewest blocks.

Success criteria

- I can change the number on a block.
- I can count squares to choose the number.
- I can use fewer blocks.

Key vocabulary

- **number**: Tells the block how many.
- **steps**: Squares the character moves.
- **fewest**: The smallest number of blocks.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (30 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Jump game (5 min): call out “frog, jump 3!” and children jump that many times.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Big moves in ScratchJr: learners follow the steps.
5 min	**Perform**	One big step, Up and over, Same moves, Is it a short way or a long way to write it?
3 min	**Probe**	Zig-zag
4 min	**Wrap-up**	Exit ticket: Show me the block that moves the cat 4 squares right.

Unplugged warm-up

Jump game (5 min): call out “frog, jump 3!” and children jump that many times.

Teaching the idea

Block parameters: the number on a motion block sets how many squares to move. Using parameters makes programs shorter (efficiency). Children count squares and enter the number.

Analogy: Hopscotch: “jump 3” instead of “jump, jump, jump”.

Live demo: Show “right 1” three times, then replace them with “right 3”. Same result!

Questions to ask

Ask	Listen for
What does the number do?	Tells how many squares.
Which is shorter: three blocks or one block with 3?	One block with 3.
How do you know the number?	Count the squares.

Differentiation

Support
Children touch and count each square on the screen before choosing the number.

Stretch
Make a path with 4 turns using only 5 blocks.

Common misconceptions

Misconception	How to fix it
A bigger number makes the cat go faster	It goes further, not faster.
Numbers only work on right arrows	Every arrow block has a number.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

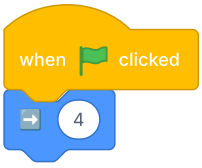
Exit ticket
Show me the block that moves the cat 4 squares right.

Homework
Play “Simon says jump 2, hop 3” with your family.

Answer key — Kitabash activities

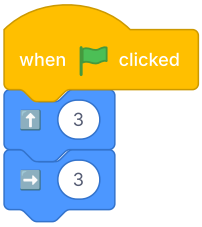
1. One big step (perform · block code)

One solution:



2. Up and over (perform · block code)

One solution:



3. Same moves (perform · multiple choice)

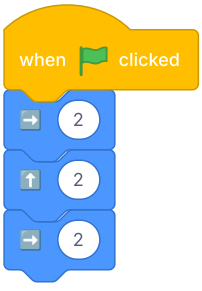
Answer: ➡️ 4

4. Is it a short way or a long way to write it? (perform · categorize)

Short: ➡️ 3, ⬆️ 2
Long: ➡️ ➡️ ➡️, ⬆️ ⬆️

5. Zig-zag (probe · block code)

One solution:



Answer key — worksheet

1. **Key words:** number = Tells the block how many. · steps = Squares the character moves. · fewest = The smallest number of blocks.

2. **Same moves:** A. ➡️ 4 — Four steps right in one block.
3. **Is it a short way or a long way to write it?:** Short: ➡️ 3, ⬆️ 2 · Long: ➡️ ➡️ ➡️, ⬆️ ⬆️
4. **Write the number:** ➡️ 4, ⬆️ 2

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.